De La Salle University
College of Computer Studies

System Administration Laboratory Manual

Group Members (LN, FN):	Quizon, Marc Lesley Khan, Sarah	Date:	**Feb 6, 2026**
Section:	S18 Group 8	Grade:	

7.0 Linux Installation

One of the variations of the Linux operating system is Ubuntu. Ubuntu is based on the Debian/GNU Linux distribution. Ubuntu is free to use Linux version without license restriction. For the purpose of this course, the Ubuntu server with long term support (LTS) variation is to be used. Aside from the Ubuntu operating system, the course also uses virtualization software to allow installation of Ubuntu without changing the host's operating system of the computer to be used.

During installation of Ubuntu Server, the installer needs information on the computer's keyboard, time zone. Another task during installation is setting up the disk partition, user account, passwords, and software packages.

7.1. Cloud-based Virtual Machine (VM)

In this section, the cloud-based virtual machine is already created. This laboratory uses the Proxmox VE system as the virtual machine server. The virtual machine settings in the server are:

Table 7-1 Virtual machine setting in cloud server

Processor	2
Memory	1 GB
Hard Disk Size	50 GB
Network Adapter	Vmbr3
CDROM	Ubuntu Server 22.04.3 ISO
Display	Auto Detect

Step 1 Server can be accessed by typing in `https://ccscloud.dlsu.edu.ph` in the URL text box of the browser. When browser prompts that website is not secure or is a risk, accept the risk and continue. This is because the server is using a self-signed certificate.

Step 2 Login to the server with the assigned username and password. Ensure that "Realm" is using "Proxmox VE authentication server". Example login page is shown in Figure 7-1.

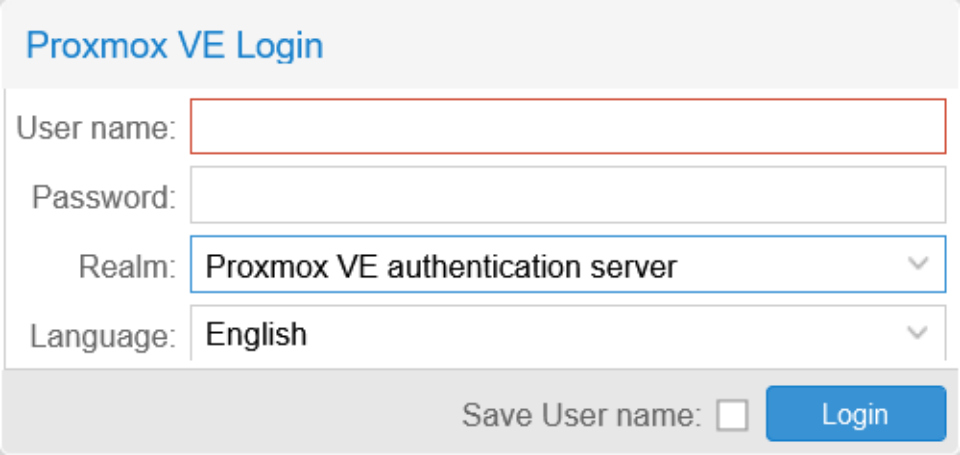


Figure 7-1 Proxmox VE login page

Step 3 Select the virtual machine and then click “Start”. This will power-up the virtual machine. Power-up button for virtual machine is shown in Figure 7-2.

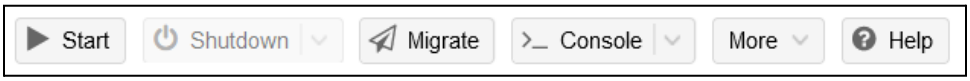


Figure 7-2 Proxmox VE user interface

Step 4 The console for the virtual machine can be accessed by using the console button shown in Figure 7-2.

7.2. Ubuntu Server Installation

Step 5 Start the virtual machine created for your Ubuntu installation. Make sure that your virtual machine boots from the CDROM with the ISO image (**ubuntu-…..iso**).

5a. During boot-up, if asked, select “**Try or Install Ubuntu**.”

Step 6 If you are asked if you want to update the installer to the latest version, select “Continue without updating”. In the language configuration page, choose the English language as shown in Figure 7-3.

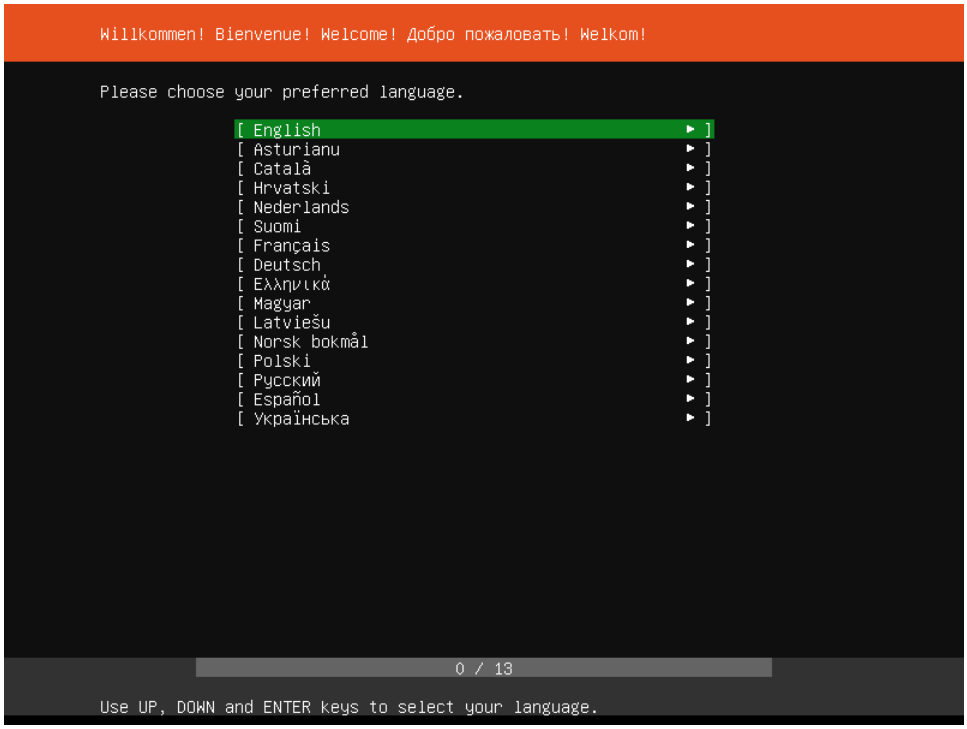


Figure 7-3 Ubuntu installation language page

Step 7 In the keyboard layout page, choose ‘English (US)’ for both the keyboard layout and variant, shown in Figure 7-4.

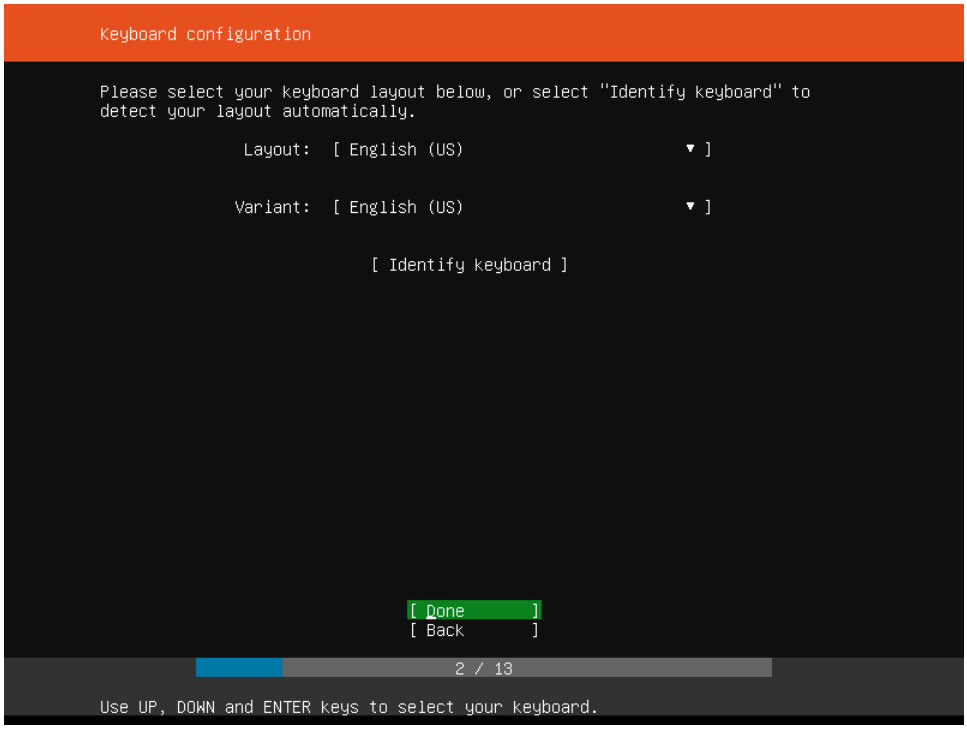


Figure 7-4 Ubuntu installation keyboard layout page

Step 8 For the type of installation, select the first Ubuntu Server option (NOT minimized). Do not check the additional options. See Figure 7-5.

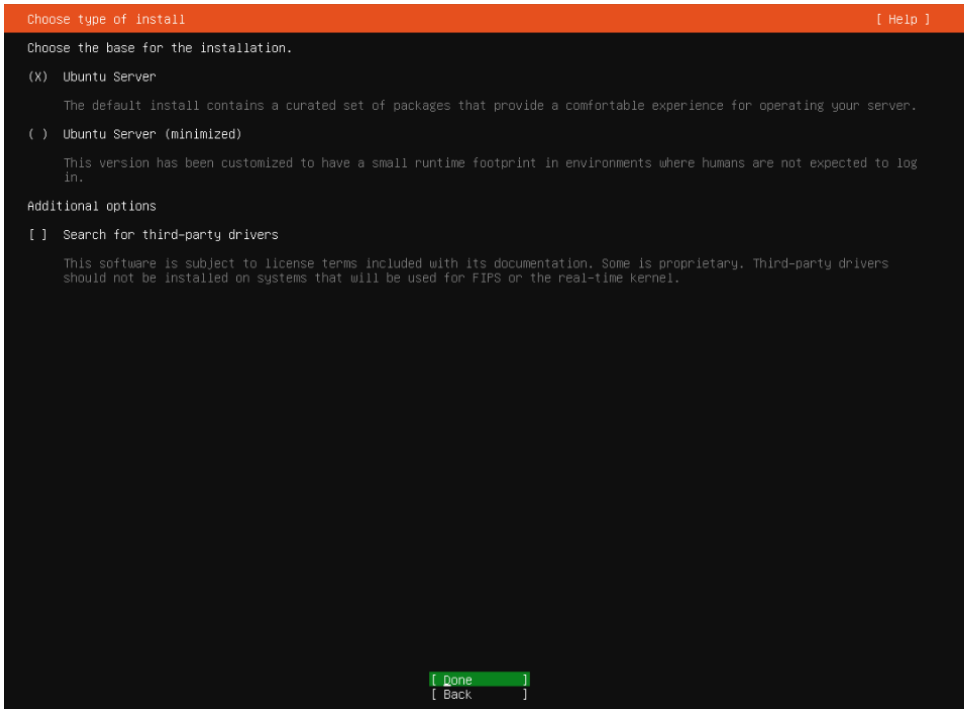


Figure 7-5 Ubuntu installation type of install page

Step 9 In the Network Connections page, leave the setting to DHPC as shown in Figure 7-6. Note that your actual output (such as network address) may differ from the provided screenshot.

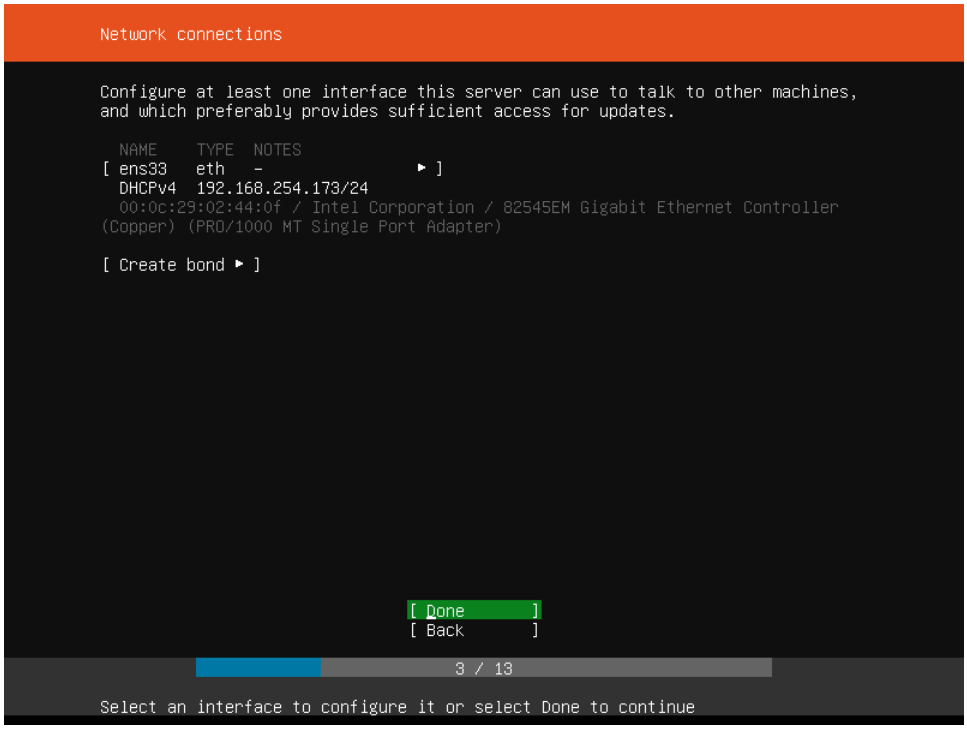
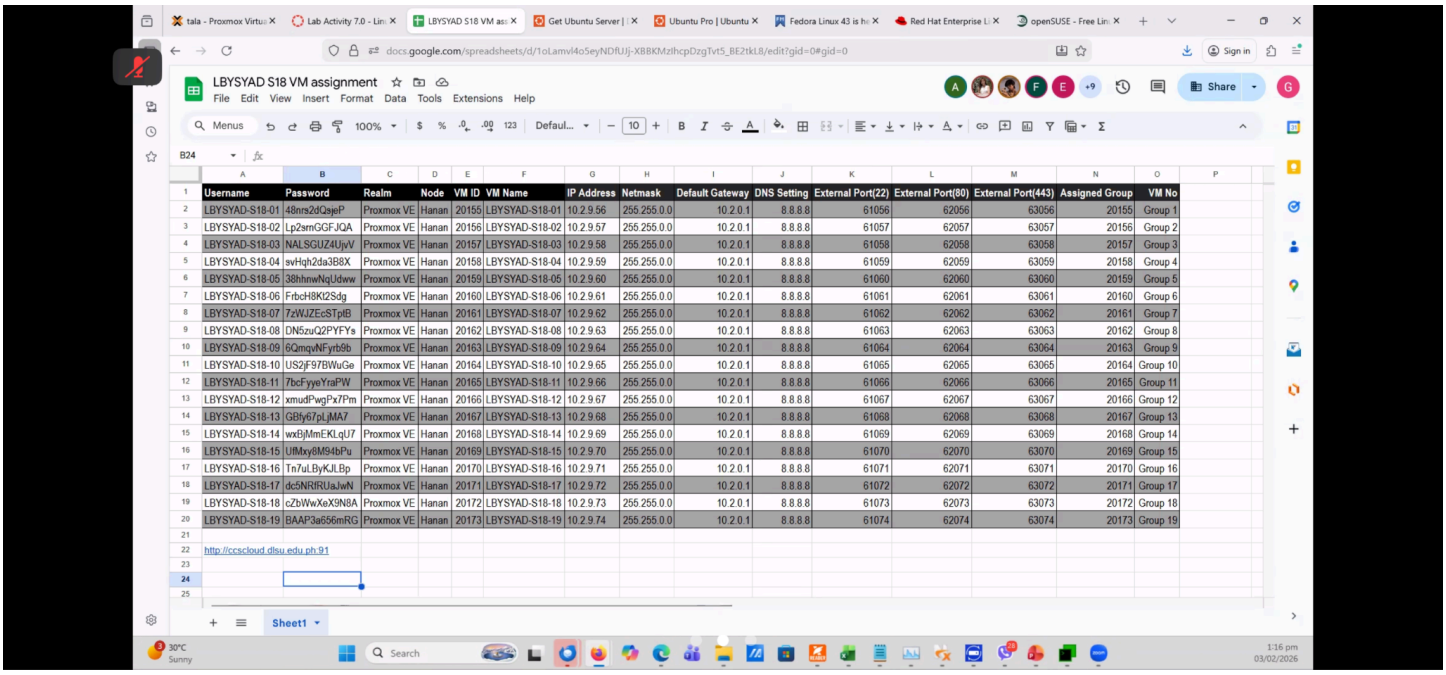


Figure 7-6 Ubuntu installation network connection page



LBYSYAD S18 VM assignment

Username	Password	Realm	Node	VM ID	VM Name	IP Address	Netmask	Default Gateway	DNS Setting	External Port(22)	External Port(80)	External Port(443)	Assigned Group	VM No
LBYSYAD-S18-01	48nrs2dQsjeP	Proxmox VE	Hanan	20155	LBYSYAD-S18-01	10.2.9.56	255.255.0.0	10.2.0.1	8.8.8.8	61056	62056	63056	20155	Group 1
LBYSYAD-S18-02	Lp2srnGGFJQA	Proxmox VE	Hanan	20156	LBYSYAD-S18-02	10.2.9.57	255.255.0.0	10.2.0.1	8.8.8.8	61057	62057	63057	20156	Group 2
LBYSYAD-S18-03	NALSGUZ4UjvV	Proxmox VE	Hanan	20157	LBYSYAD-S18-03	10.2.9.58	255.255.0.0	10.2.0.1	8.8.8.8	61058	62058	63058	20157	Group 3
LBYSYAD-S18-04	svHqh2da3B8X	Proxmox VE	Hanan	20158	LBYSYAD-S18-04	10.2.9.59	255.255.0.0	10.2.0.1	8.8.8.8	61059	62059	63059	20158	Group 4
LBYSYAD-S18-05	38hhnwNqUdww	Proxmox VE	Hanan	20159	LBYSYAD-S18-05	10.2.9.60	255.255.0.0	10.2.0.1	8.8.8.8	61060	62060	63060	20159	Group 5
LBYSYAD-S18-06	FrbcH8Kl2Sdg	Proxmox VE	Hanan	20160	LBYSYAD-S18-06	10.2.9.61	255.255.0.0	10.2.0.1	8.8.8.8	61061	62061	63061	20160	Group 6
LBYSYAD-S18-07	7zWJZEcSTptB	Proxmox VE	Hanan	20161	LBYSYAD-S18-07	10.2.9.62	255.255.0.0	10.2.0.1	8.8.8.8	61062	62062	63062	20161	Group 7
LBYSYAD-S18-08	DN5zuQ2PYFYs	Proxmox VE	Hanan	20162	LBYSYAD-S18-08	10.2.9.63	255.255.0.0	10.2.0.1	8.8.8.8	61063	62063	63063	20162	Group 8
LBYSYAD-S18-09	6QmqvNFyrb9b	Proxmox VE	Hanan	20163	LBYSYAD-S18-09	10.2.9.64	255.255.0.0	10.2.0.1	8.8.8.8	61064	62064	63064	20163	Group 9
LBYSYAD-S18-10	US2jF97BWuGe	Proxmox VE	Hanan	20164	LBYSYAD-S18-10	10.2.9.65	255.255.0.0	10.2.0.1	8.8.8.8	61065	62065	63065	20164	Group 10
LBYSYAD-S18-11	7bcFyyeYraPW	Proxmox VE	Hanan	20165	LBYSYAD-S18-11	10.2.9.66	255.255.0.0	10.2.0.1	8.8.8.8	61066	62066	63066	20165	Group 11
LBYSYAD-S18-12	xmudPwgPx7Pm	Proxmox VE	Hanan	20166	LBYSYAD-S18-12	10.2.9.67	255.255.0.0	10.2.0.1	8.8.8.8	61067	62067	63067	20166	Group 12
LBYSYAD-S18-13	GBfy67pLjMA7	Proxmox VE	Hanan	20167	LBYSYAD-S18-13	10.2.9.68	255.255.0.0	10.2.0.1	8.8.8.8	61068	62068	63068	20167	Group 13
LBYSYAD-S18-14	wxBjMmEKLqU7	Proxmox VE	Hanan	20168	LBYSYAD-S18-14	10.2.9.69	255.255.0.0	10.2.0.1	8.8.8.8	61069	62069	63069	20168	Group 14
LBYSYAD-S18-15	UfMxy8M94bPu	Proxmox VE	Hanan	20169	LBYSYAD-S18-15	10.2.9.70	255.255.0.0	10.2.0.1	8.8.8.8	61070	62070	63070	20169	Group 15
LBYSYAD-S18-16	Tn7uLByKJLBp	Proxmox VE	Hanan	20170	LBYSYAD-S18-16	10.2.9.71	255.255.0.0	10.2.0.1	8.8.8.8	61071	62071	63071	20170	Group 16
LBYSYAD-S18-17	dc5NRfRUaJwN	Proxmox VE	Hanan	20171	LBYSYAD-S18-17	10.2.9.72	255.255.0.0	10.2.0.1	8.8.8.8	61072	62072	63072	20171	Group 17
LBYSYAD-S18-18	cZbWwXeX9N8A	Proxmox VE	Hanan	20172	LBYSYAD-S18-18	10.2.9.73	255.255.0.0	10.2.0.1	8.8.8.8	61073	62073	63073	20172	Group 18
LBYSYAD-S18-19	BAAP3a656mRG	Proxmox VE	Hanan	20173	LBYSYAD-S18-19	10.2.9.74	255.255.0.0	10.2.0.1	8.8.8.8	61074	62074	63074	20173	Group 19

http://ccscloud.dlsu.edu.ph:91

Step 10 In the Configure Proxy page, do not input any proxy server as shown in Figure 7-7.

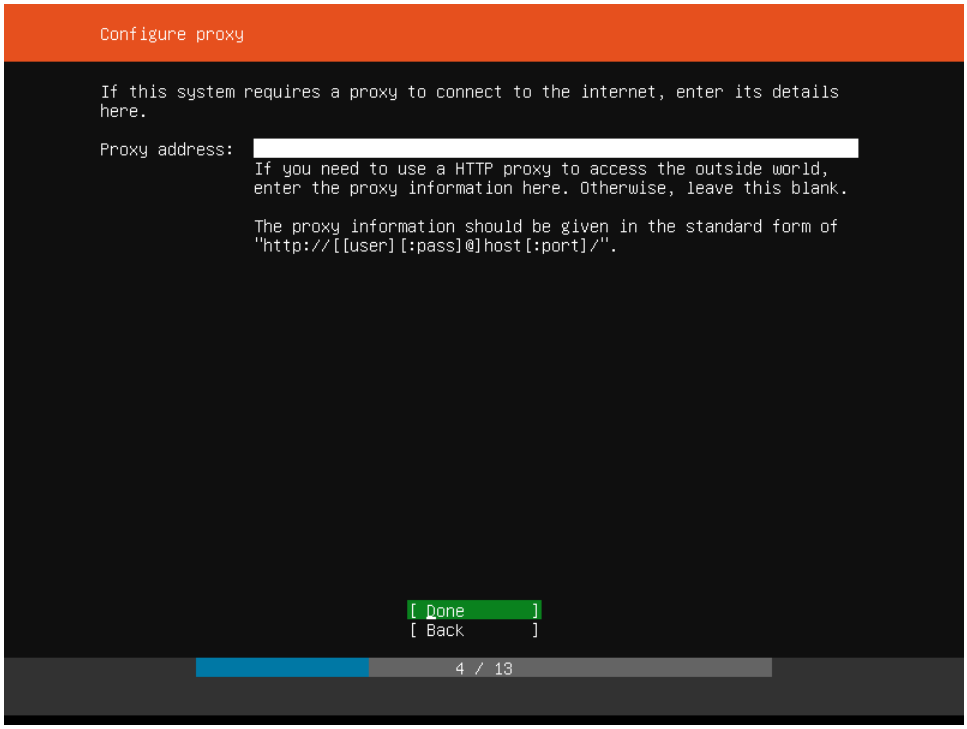


Figure 7-7 Ubuntu installation proxy configuration page

Step 11 In the Configure Archive Mirror page, accept the default and press **[DONE]** as shown in Figure 7-8.

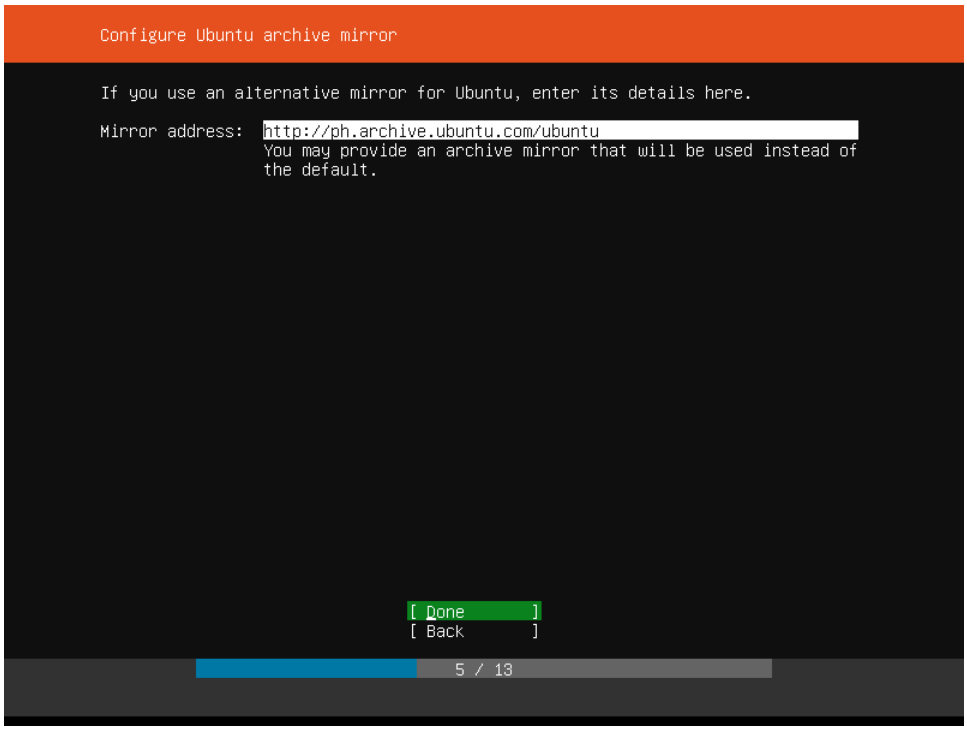


Figure 7-8 Ubuntu installation archive mirror configuration page

Step 12 In the Guided storage configuration page, choose the "**Use an entire disk**" option as shown in Figure 7-9.

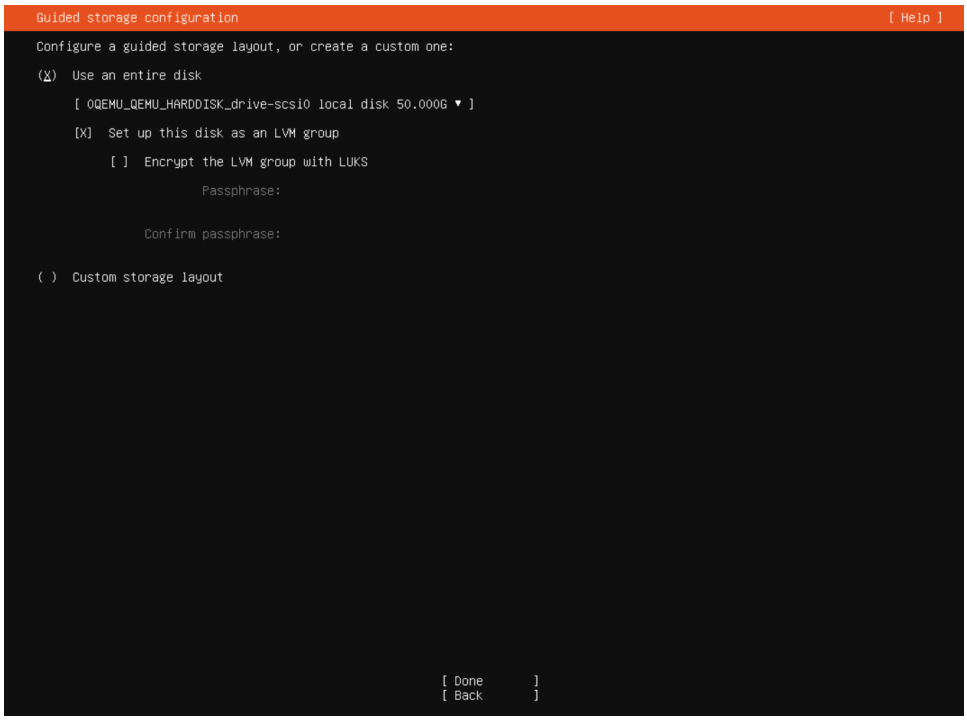


Figure 7-9 Ubuntu installation Guided storage configuration page

12a. If asked which disk to use, choose ‘dev/sda’ or the default disk listed on the page as shown in Figure 7-10.

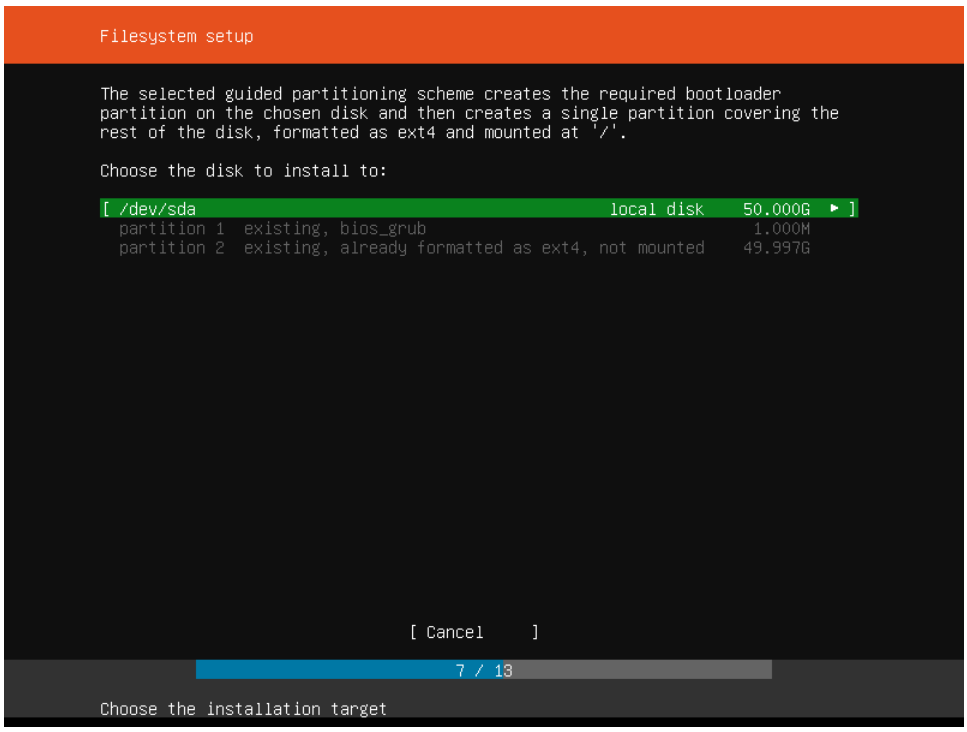


Figure 7-10 Ubuntu installation filesystem setup for disk

12b. When the filesystem setup summary is presented, press **[DONE]** to accept the filesystem configuration as shown in Figure 7-11.

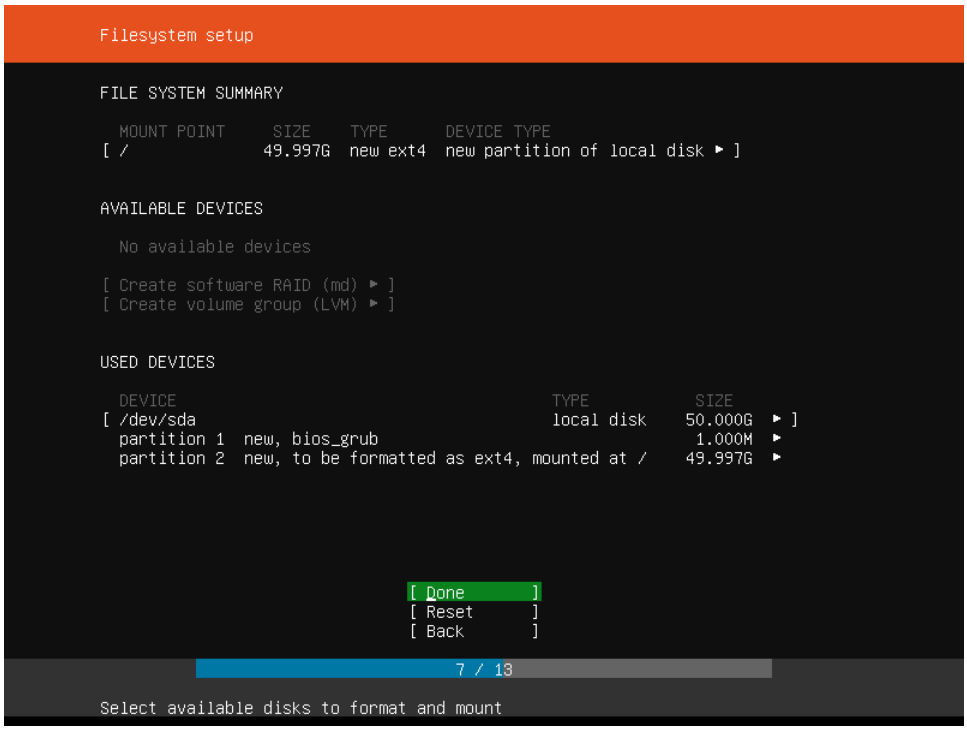


Figure 7-11 Ubuntu installation filesystem setup summary

12c. In this step, Ubuntu installation will reconfirm the action as it is a destructive step. Choose the “Continue” option to proceed as shown in Figure 7-12.

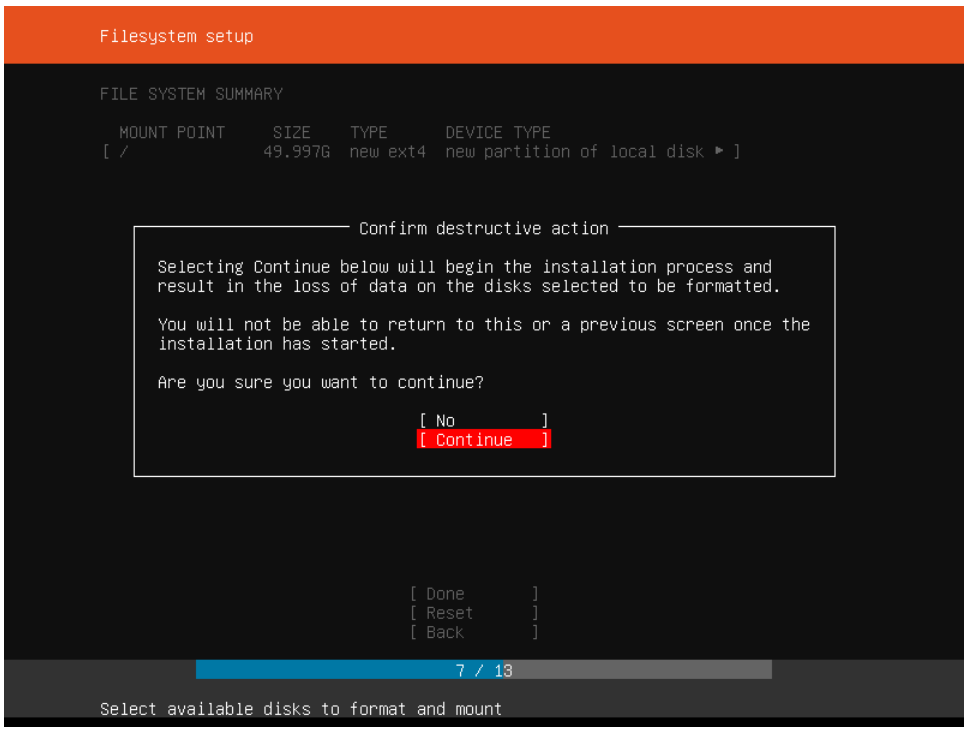


Figure 7-12 Ubuntu installation reconfirmation for filesystem setup

Step 13 In the Profile Setup page, input the following:

Name	<Any group member’s name>
Server name	lbysyad-<section>-<VMnumber> or issysad-<section>-<VMnumber>
Username	administrator
Password	DLSU1234!

Press **[DONE]** when all the required information has been inputted similar to what is shown in Figure 7-13.

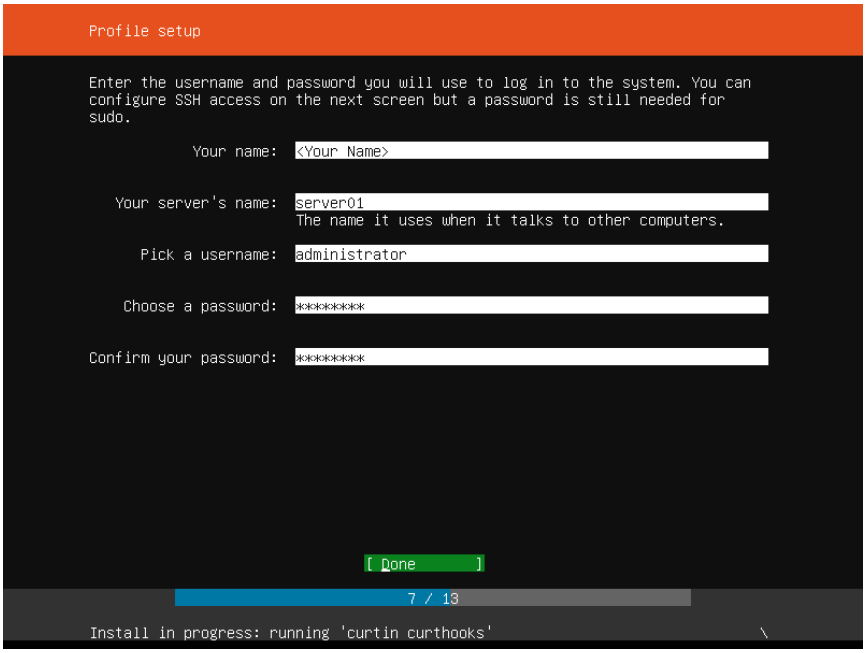


Figure 7-13 Ubuntu installation profile setup

Step 14 If asked if you want to upgrade to Ubuntu Pro, select "**Skip for now**."

Step 15 In the SSH setup, **choose to install the OpenSSH** server. Use the space bar to toggle the option. Do not import SSH identity. Press **[DONE]** as shown in Figure 7-14.

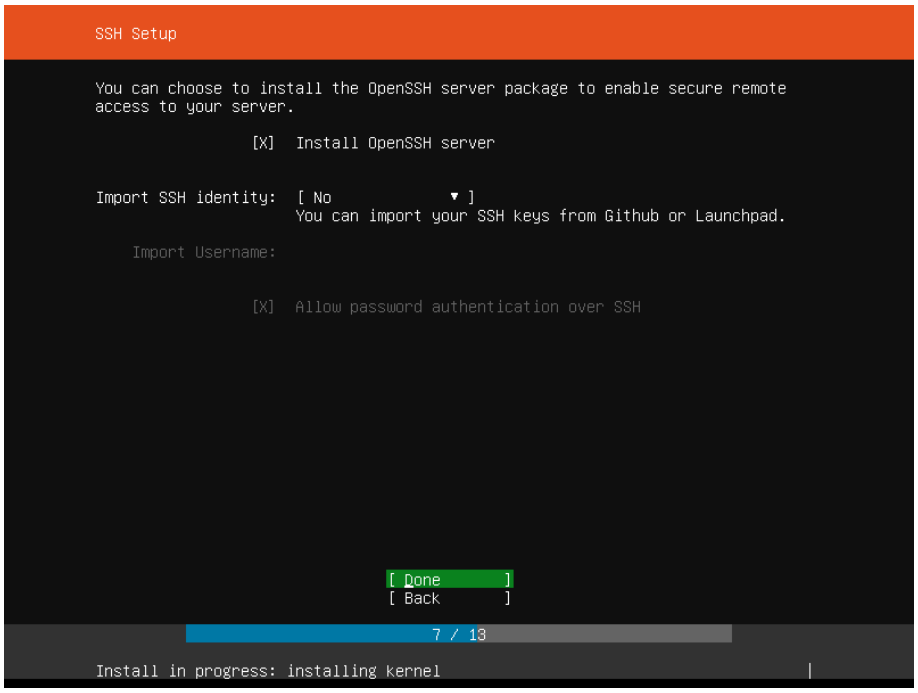


Figure 7-14 Ubuntu installation SSH setup

Step 16 In the Featured Server Snaps page, do not choose any software package to install and press **[DONE]** as shown in Figure 7-15.

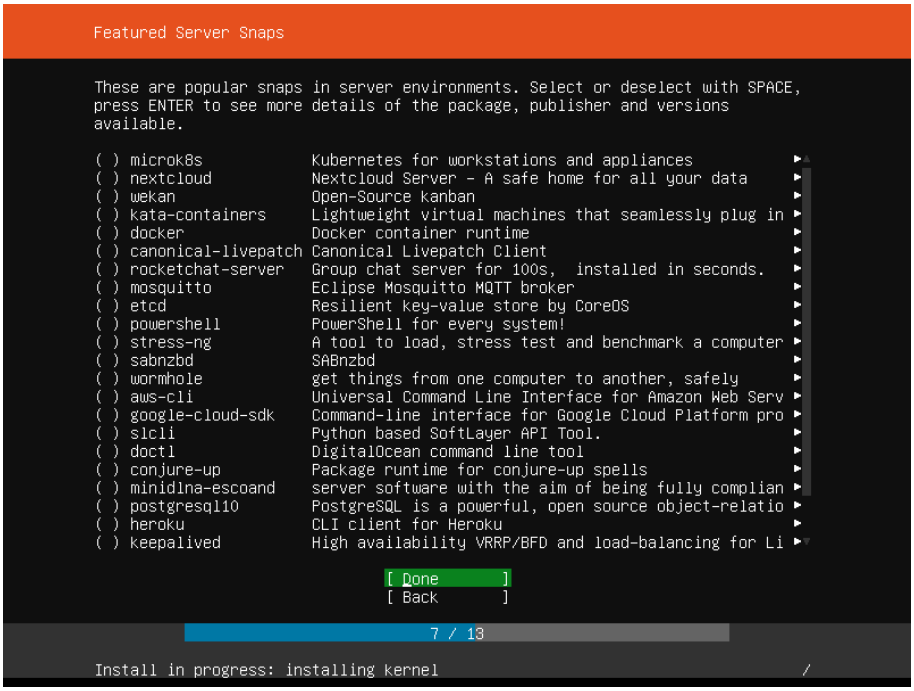


Figure 7-15 Ubuntu installation software package installation

Step 17 If there is no problem in the installation setup, the kernel and other packages will be installed as shown in Figure 7-16.

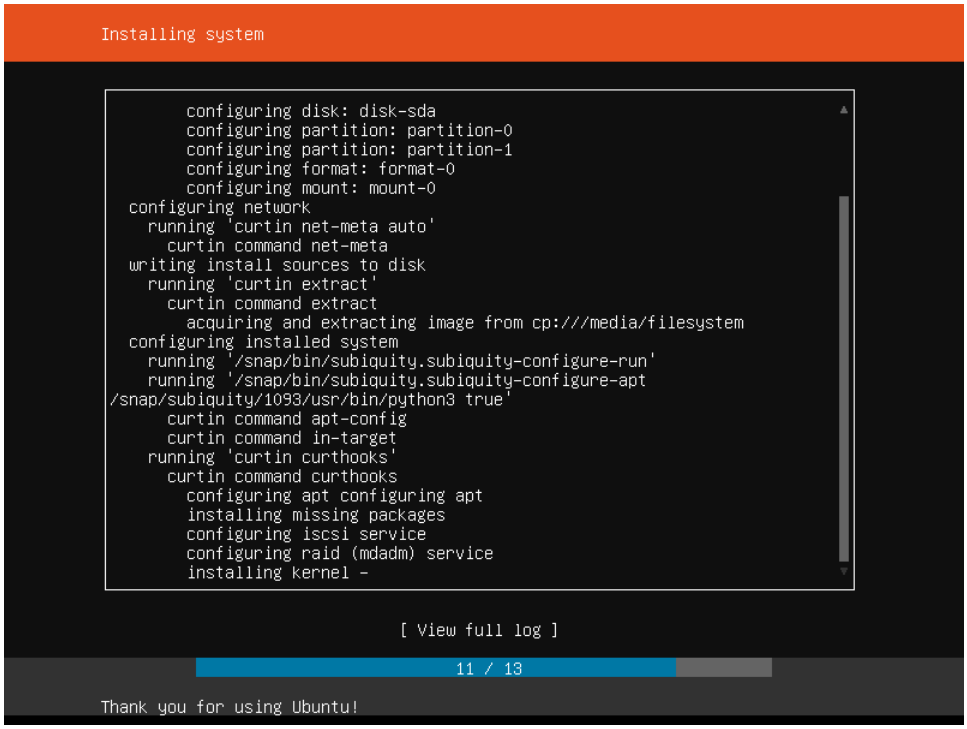


Figure 7-16 Ubuntu installation of kernel and other packages

Step 18 During the Installation Update phase, cancel the update and reboot as shown in Figure 7-17.

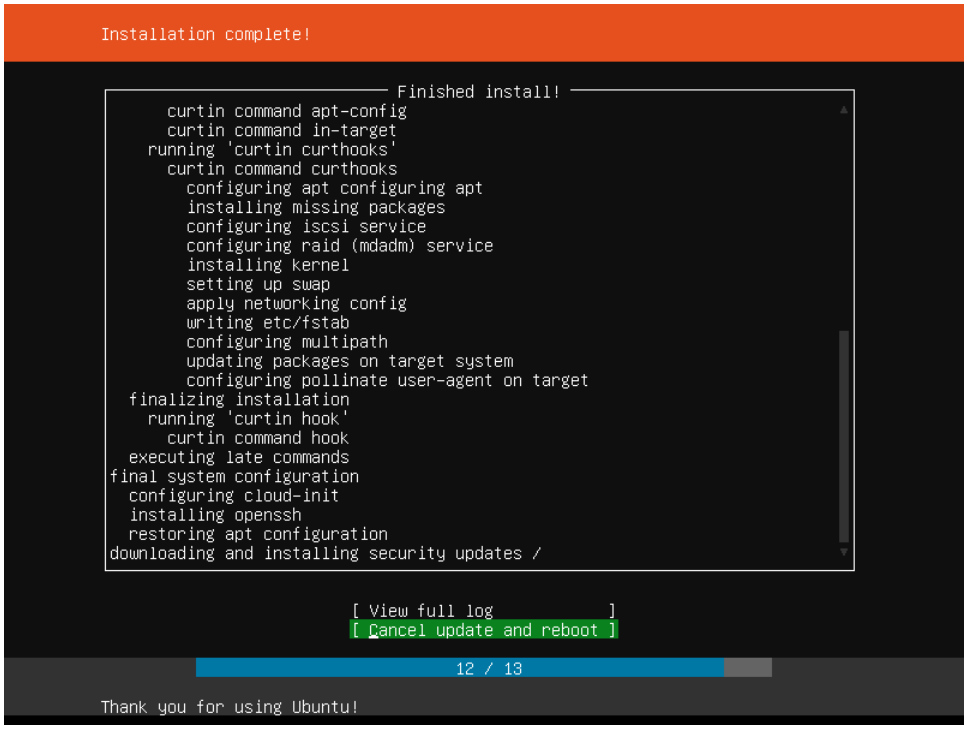


Figure 7-17 Ubuntu installation update

Step 19 After rebooting, if you encounter a cdrom error similar to what is shown in Figure 7-18, just press the enter key to continue booting. To avoid this error in the future, before booting your virtual machine, go to Proxmox Hardware options and edit the CD/DVD drive to not use any media (see Figure 7-19).

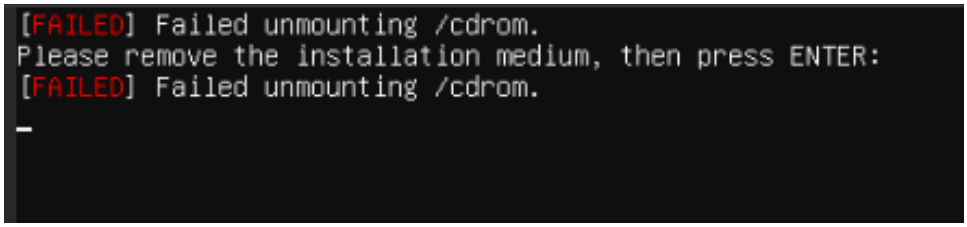


Figure 7-18 Unmounting error message

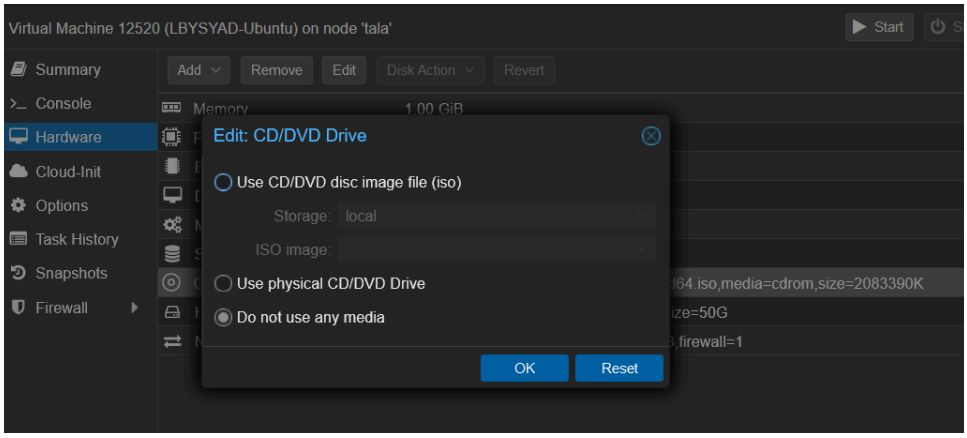


Figure 7-19 Removing the virtual installation disc in Proxmox

Step 20 If no errors are encountered after the system reboots, the server should boot as shown Figure 7-20. Note that your output may differ from the given screenshot.

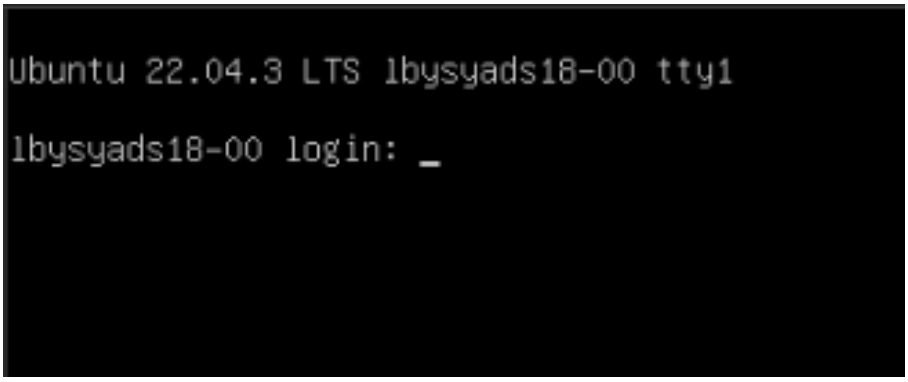


Figure 7-20 Running Ubuntu system

20a. What is the computer name of the installed server? Is it the same as the name you provided during installation?

lbysyad-s18-20162

20b. Try to login with the initial username and password. Were you able to login? Why or why not?

Oh yes indeed. We were able to login because we set up the profile correctly

20c. What is the private IP address and netmask of your recently installed server as seen on the interface?

The address is 10.2.9.63/16

20d. On the prompt, type in the command '`cat /proc/cpuinfo`'. How many CPUs are there?

There is 1 CPU/Processor.

7.3. Conclusion/Reflection

Using the assigned port number to your machine, try to SSH into it using PuTTY or the SSH command via Command Prompt/Terminal. Were you able to login? Why or why not? Justify your answer using your knowledge of computer networking concepts.

Yes, we can log in using SSH. The port was redirected to a new number 61063.

Note: Always shut down the server after completing the activity. Shut down the server by typing in the command '`shutdown -h now`'. This will shut down the server properly. The shutdown command will be explained later in the succeeding lab activity.